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Performance-oriented system digital twin: a Bayesian network-based framework to customize Sub-digital twins

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ABSTRACT

Bayesian network (BN) has emerged as a fundamental mathematical tool for implementing digital twins (DTs) in complex systems. Despite substantial progress in building DTs via BNs, challenges remain in storage and computational efficiency of BNs as complexity of the system scales. This study develops a scalable BN-based framework to customize sub-DTs from system digital twins (SDTs) based on specified performance measures, termed performance-oriented SDT. The SDT uses mutable knowledge graphs (KGs) to represent complex systems, enabling the capture of statistical correlations across different sub-systems while providing flexibility to model the dynamic evolution and transformation of infrastructure systems. To support event-responsive decision under any scenarios, a generic algorithm is proposed to create custom sub-DTs from KGs for selected performance measures and available data sources. For proof-of-concept, a hypothetical graph is first presented as a numerical example to demonstrate the versability and flexibility of the proposed model/algorithm for DT-supported management of infrastructure systems. Then model/algorithm is applied to the bridge network in Miami-Dade County: results show that a custom SDT with the size of 1% of the original KG can be created for practical use of the approach that significantly improves the computational efficiency in DT applications.

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1. Introduction

Management of infrastructure systems is critical for serving the welfare of society. With recent advances in computational power, information science, and communication technologies, digital twin (DT) has emerged as a promising decision support tool gaining increasing popularity for infrastructure management (Broo et al., 2022; Errandonea et al., 2020; Pregnolato et al., 2022). Generally, DT is a digital or virtual representation that can mirror the current states and predict the future performance of a physical asset or system. Compared to a traditional model such as building information modeling (BIM) and finite element models, DT distinguishes itself by harvesting and integrating big data to automatically and continuously update the evolving virtual model to reflect and predict the conditions and performance of the physical asset over a long period time (Kunzer et al., 2022; Vieira et al., 2022; Wagg et al., 2020). For DT applications in infrastructure management, existing studies have shown the benefits of inspections and structural health monitoring through value of information analysis (Andriotis et al., 2021; Zhang et al., 2021). It is broadly expected that DT-based approaches can significantly reduce the life-cycle cost due to its ability to synchronize between the physical and virtual infrastructure systems and to enable predictive management/maintenance.

There have been good headways made in DT applications within Architecture, Engineering, Construction, and Operations (AECO) industries, particularly in waves of efforts toward modeling smart cities. Various implementations of DT have been explored across different infrastructure sectors such as water distribution systems (Conejos Fuertes et al., 2020), bridges (Lu & Brilakis, 2019; Pregnolato et al., 2022; Shim et al., 2019), and buildings (Francisco et al., 2020; Lu et al., 2020; Petri et al., 2023). Taking bridges as an example, notable DT efforts have concentrated on enhancing life-cycle management, such as integrating BIM with 3D scanning to improve bridge maintenance process (Shim et al., 2019) and utilizing Bayesian inference of the deterioration parameters for real-time updating of the fatigue evolution process (Jiang et al., 2023). A suite of models involving measurements, simulation, 3D modeling, and validation have been developed to create a cyber-physical bridge DT during load testing (Chacón et al., 2024). Additionally, Hagen and Andersen (2024) illustrated the value of DTs in emergency response and management through an alarm system based on a bridge DT. These studies represent a good sample of existing attempts to implement DT concepts, where they emphasize the integration of modeling enabling technologies such as geometry,

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simulations, measurements, and other data-driven techniques (Thelen et al., 2022).

In the meantime, Bayesian network (BN) has emerged as an essential component within the DT framework as its capabilities align closely with the core principles and objectives of DT technology (VanDerHorn & Mahadevan, 2021). Specifically, DTs for complex systems are expected to achieve three core functions: (a) modeling complex correlations and interdependencies, (b) assimilating heterogeneous data streams from real-world observations, and (c) performing probabilistic inference to support decision-making. BNs inherently support these functions through their unique capabilities as follows. (a) Graphical representation of dependencies: BNs provide a graphical formulation where nodes represent components of the system and edges denote probabilistic dependencies between them, facilitating the modeling of complex correlations across various scales within a system. (b) Data assimilation: BNs enable seamless integration of heterogeneous data streams by specifying data sources as nodes within the network, allowing for the assimilation of multiple types of evidence within a unified analytical framework. (c) Bayesian updating: BNs enable probabilistic reasoning by updating the beliefs about variables of interest based on available evidence, supporting informed decision-making for system operation, management, and maintenance. Given these capabilities, BNs have emerged as a pivotal component in the development and implementation of DTs.

Extensive studies across various research areas have employed BNs as the fundamental mathematical tools for DTs. For example, Kapteyn et al. (2021) proposed a DT framework for an unmanned aerial vehicle (UAV) using a dynamic Bayesian network (DBN) with the addition of decision nodes. In this framework, information flows from physical assets are assimilated at the data node to update the digital state node. The updated digital state can inform the decision node, facilitating a feedback loop between the digital and physical spaces. Similarly, Torzoni et al. (2024) developed a DBN-based DT framework for monitoring and managing civil infrastructures. Sequential sensor observations are processed using deep learning models to predict the health states of physical structures. These predictions are used to update the digital state node via Bayesian inference, thereby informing decision nodes for optimized maintenance and management actions.

Additionally, Braik and Koliou (2023, 2024) applied DBNs to develop DTs for electric power infrastructure systems. DTs capture complex correlations within the system and interdependencies between the system and road networks through the graphical representation of BNs. Various sources of evidence are integrated at data nodes and then Bayesian inference is performed to update the performance state of system elements. Geng et al. (2025) developed a system digital twin (SDT) to monitor the risk of bridge network in Miami-Dade County. The SDT integrates data from three interconnected systems and incorporates statistical correlations for risk updating. These studies collectively affirm that BN is a valid and widely used mathematical approach within the scope of DT. However, significant challenges remain, particularly in terms of the storage and computational efficiency of BNs as complexity of the system scales. For example, the SDT representing the bridge network in Miami-Dade County consists of 6,478 nodes and 10,584 edges, requiring 97,484.8 MB of storage memory (Geng et al., 2025). This poses a significant computational challenge in practical implementations of the SDT.

To address the computational challenges of large-scale SDTs, a framework is developed to customize sub-DTs from the SDT based on selected performance measures and available data sources, termed as performance-oriented SDT. It aims to address three challenges in developing SDT for interdependent infrastructure systems:

I. Statistical correlations and interdependencies at different levels of the complex systems need to be accounted for. These correlations and dependencies exist within individual structures (e.g. multiple failure modes of a structure), between structures (e.g. bridges in a transportation network under spatially distributed hazards), and across infrastructure systems (e.g. water and power distribution systems with interdependencies).
II. The transformation and evolution of the infrastructure systems over time need to be modeled, monitored, and managed dynamically and adaptively. For example, the incremental development of smart infrastructure can be achieved by installing new sensors and constructing new infrastructure facilities, while some existing devices and facilities may be demolished after reaching their service lives.
III. There exist event-responsive reasonings and various decision-making scenarios for modeling and managing a complex system. Numerous what-if scenarios and questions need to be modeled/answered often in the following format: given certain observations $\boldsymbol{O}$, how can event A affect event B or variable C? For example, given multiple data sources in modeling transportation systems, how can a 10% annual decrease in emission factors of vehicles affect future carbon dioxide emissions? Therefore, for stakeholders, the SDT models should be flexible to account for any combinations of events, objectives, performance measures, and various data sources. Overall, for the modeling and management of complex infrastructure systems, it is essential to develop a scalable SDT framework to tackle these three challenges.

The performance-oriented SDT supports a user-centric decision support system architecture that can be easily customized according to the specific needs of a decision-maker. It is built on the premise that a mutable knowledge graph (KG) representing the targeted infrastructure system of interest is stored in the performance-oriented SDT. Then based on the performance measure (PM) and available data specified by the decision-maker, the proposed algorithm can automatically generate a sub-DT from the stored KG. The sub-DT does not imply a structural decomposition of a DT

for a single bridge but rather a functional partitioning of a large-scale SDT tailored to specific objectives. By pruning nodes and edges irrelevant to the targeted performance measures, the proposed algorithm can substantially reduce the size of the graph, thus improving the computational efficiency of the SDT. Particularly, Bayesian inference enables the proposed SDT to address challenge I, while the mutable KG in BN can tackle challenge II. Finally, the proposed algorithm for automatic generation of custom DTs offers the performance-oriented SDT the flexibility and adaptability for overcoming challenge III.

In the following sections, through a hypothetical graph representing abstract interdependent infrastructure systems, the conceptual framework of SDT is presented and detailed algorithms are developed in support of BN modeling of SDT. Specifically, the performance-oriented SDT is first described in terms of how it can create custom trunks consistent with intuition from a hypothetical infrastructure systems graph based on users' inputs for their decision-making scenarios. Extending one of the decision scenarios, a data-driven infrastructure system reliability management case is then analyzed to illustrate how the proposed method can transform the BN trunk to an SDT that can incorporate newly available data for posterior system reliability. Results from such a scenario analysis case demonstrate that the performance-oriented SDT can serve as a general and flexible framework with broad application potential for infrastructure systems modeling, monitoring, and management. In the end, a further in-depth case study of a bridge network in Miami-Dade County is conducted to demonstrate how the proposed framework can be implemented for practical applications. Using the algorithms developed in the framework, the automatically generated custom SDT has a size that is approximately 1% of the original KG that significantly improves the computational and storage efficiency in DT modeling of numerous infrastructure systems management scenarios.

2. Bayesian network for system digital twin

This section first introduces basic concepts of Bayesian network (BN). Relevant studies are then presented to explain why BN is well suited for establishing system digital twin (SDT) of infrastructure systems. This leads to an algorithm design proposed for the performance-oriented SDT to limit the size of the BN. BN presents a graphical model capable of reasoning, probabilistic analysis, and Bayesian inference. It is a directed acyclic graph $\mathcal{G}_d(\boldsymbol{V},\boldsymbol{E})$ consisting of a set of nodes $\boldsymbol{V}$ and a set of edges $\boldsymbol{E}$ that represent random variables and statistical dependencies, respectively (Marcot & Penman, 2019). Consider an example BN in Figure 1. X_1 and X_2 are root nodes, while X_3, X_4, and X_5 are child nodes, all random variables. X_1 and X_2 are described by marginal probability distributions $P(X_1)$ and $P(X_2)$, respectively. X_4, a child node of X_1 and X_2, is described by a conditional probability distribution $P(X_4|X_1,X_2)$. Similarly, X_3 and X_5 are described by conditional probability distributions $P(X_3|X_1)$ and $P(X_5|X_3,X_4)$, respectively. With the marginal and conditional probability distributions, the joint distribution is represented by the BN as:

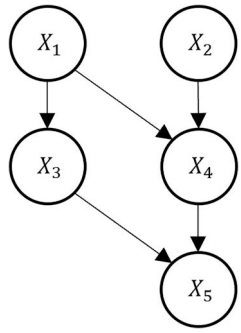


Figure 1. Example BN with five nodes.

$$P(X_1,X_2,X_3,X_4,X_5) = P(X_1)P(X_2)P(X_3|X_1)P(X_4|X_1,X_2)P(X_5|X_3,X_4) \quad (1)$$

Constructing a BN usually involves (a) establishing the graph and (2) eliciting the marginal and conditional probability distributions. The first step has to do with the system one wants to model, while the second step is a model fitting problem using data. With the BN, Bayesian updating can be efficiently conducted on a set of variables of interests based on the observations of another set of variables. Bayesian updating, also known as Bayesian inference, is a method for updating the probability distributions of variables given observations of certain other variables (Koller & Friedman, 2009). It follows Bayes' theorem, which states that the posterior probability is proportional to the prior probability multiplied by the likelihood of the observed data. Techniques such as exact inference algorithms (e.g. variable elimination, junction tree) and approximate inference methods (e.g. Monte Carlo sampling, variational inference) are employed to perform Bayesian updating efficiently. For more detailed information on basic concepts, BN construction, and Bayesian inference, readers are referred to Bensi (2010).

BN is suitable for the development of SDT for two reasons. First, as mentioned previously, an important characteristic of DT is to leverage data for better characterization of the states of physical infrastructure assets and/or systems. Data fusion methods for DT modeling based on BN and its variants (Groden & Collette, 2017; Kapteyn et al., 2021; Zhu & Collette, 2017) have been developed. One advantage of BN is that it can efficiently update variables of interest (e.g. the conditions of a bridge) when data is partially or completely available from any parts of a system. Besides, BN is well-suited for probabilistic analysis and decision-making in infrastructure systems management because it can model the statistical correlations within the analysis of a structure, a facility, or an infrastructure system and the interdependencies across infrastructure systems (Bensi, 2010). Existing studies have applied BN to reliability and/or risk assessment of single structures such as bridges (Gehl & D'ayala, 2016) and nuclear plants (Segarra et al., 2021), individual infrastructure systems such as transportation networks (Byun et al., 2019; Byun & D'Ayala, 2022; Gehl &

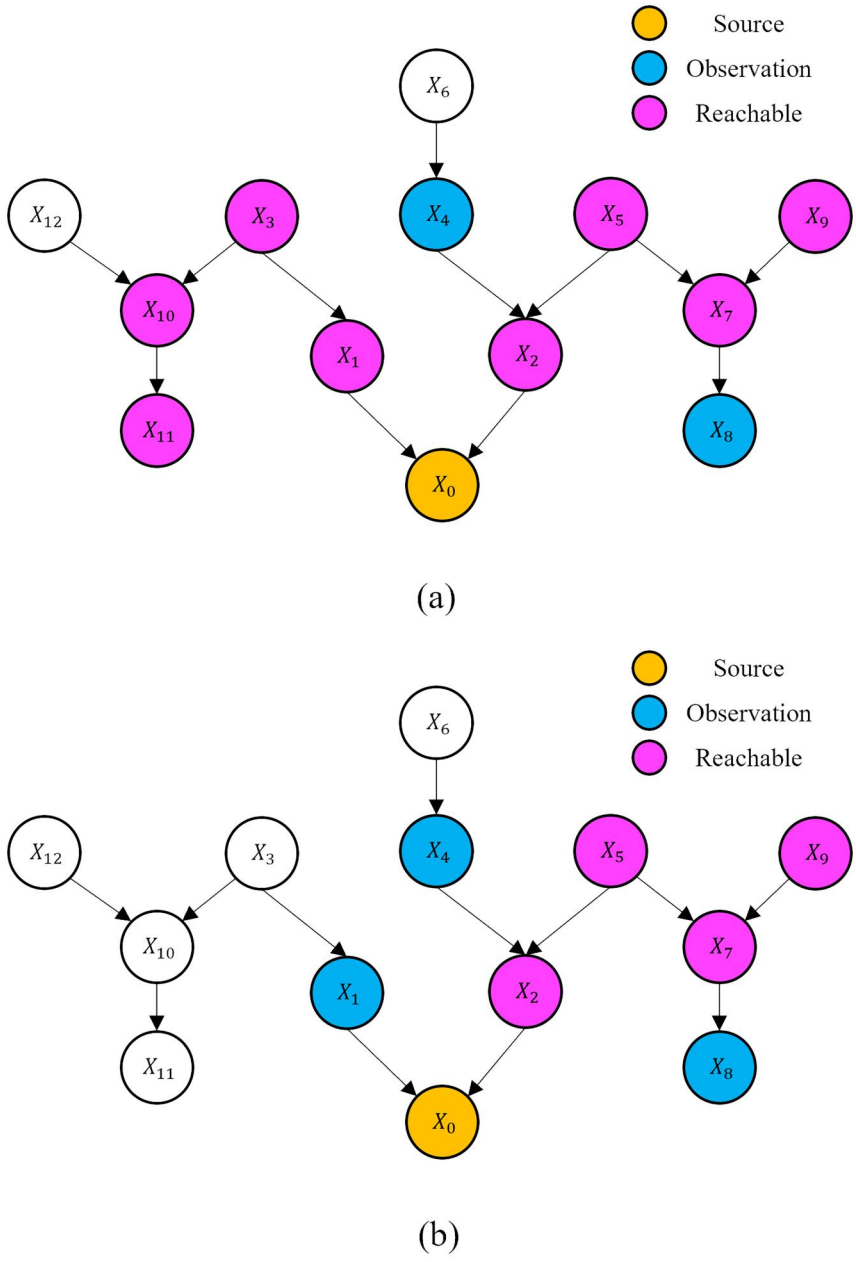


Figure 2. Illustration of the *Reachable* algorithm.

D'ayala, 2018) and power distribution systems (Tien & Der Kiureghian, 2017), and interdependent infrastructure systems including water, gas, and power networks (Applegate & Tien, 2019; Johansen & Tien, 2018).

In the application of BN for SDTs, it is critical to create a BN with a limited size. When the system becomes increasingly large and complicated, the efficiency of constructing a BN and performing inference can be compromised. Several methods have been proposed to simplify the graphical structure (Bensi et al., 2011, 2013; Straub & Der Kiureghian, 2010) and to improve efficiency in storing the conditional probability tables for inference (Byun et al., 2019; Byun & Song, 2021; Tien & Der Kiureghian, 2016, 2017). Further, reducing the size of a BN in infrastructure systems models usually improves computational efficiency on top of the achievements by the proposed methods. Therefore, it is preferable to be selective in choosing relevant nodes and links to build an SDT.

To this end, an algorithm based on the *Reachable* algorithm is proposed to create a BN at a minimum size. The inputs to the *Reachable* algorithm are the graph, the source node, and the observation nodes, while the outputs are the reachable nodes. Consider the graph with 12 nodes in Figure 2. The basic idea of the *Reachable* algorithm is to identify the reachable nodes (i.e. purple nodes) that are correlated with the source node (i.e. orange node), based on the given observation nodes (i.e. blue nodes). This algorithm is relevant to the development of SDT because the source node can be the variables of interest (e.g. seismic risk of a power distribution system), while the observation nodes represent the variables with observed data. After the algorithm identifies the reachable nodes in the graph, the reachable

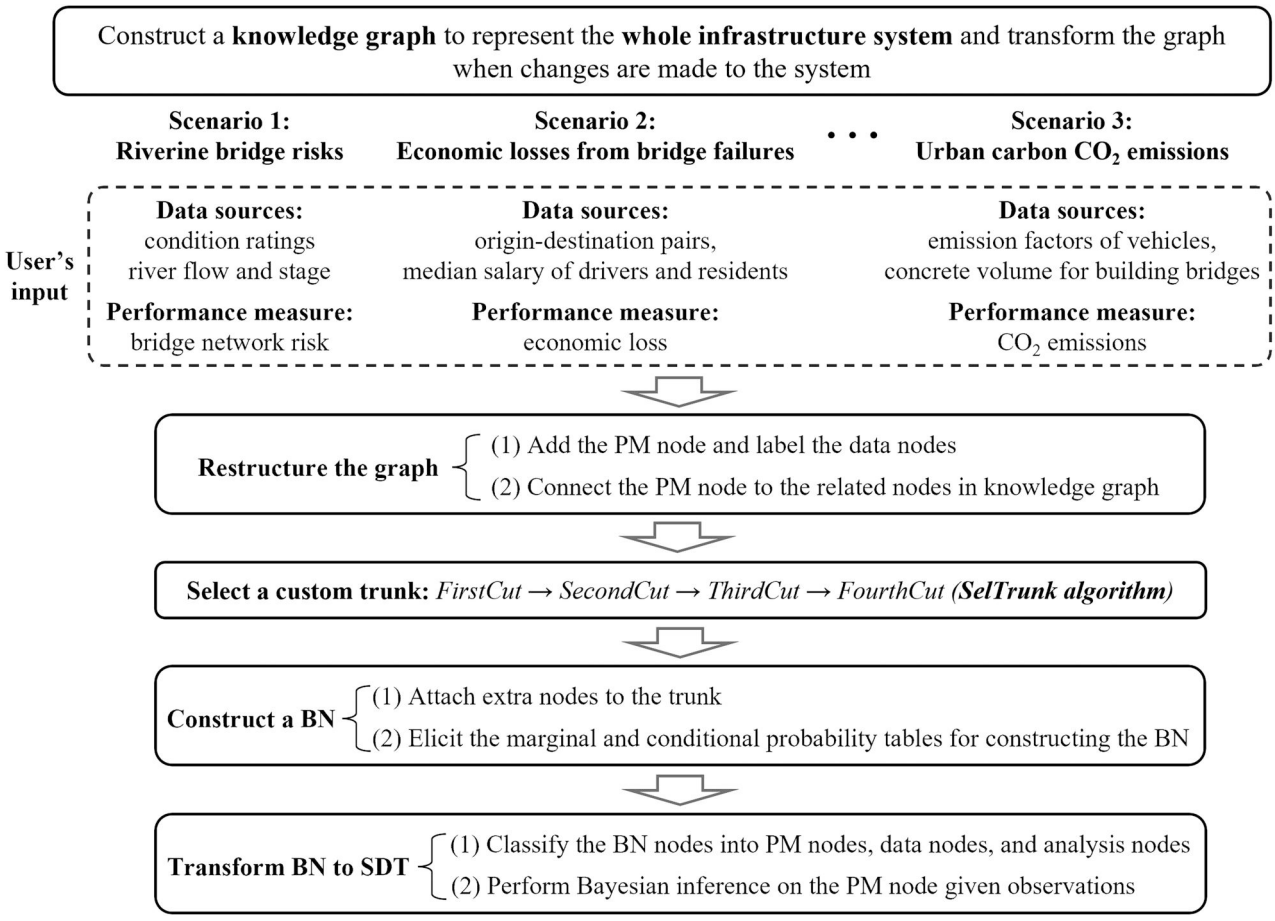


Figure 3. Architecture of performance-oriented SDT: flowchart and procedure to customize SDTs based on users' inputs.

nodes, observation nodes, and source node will be retained, while the remaining nodes that have no impacts on the source node can be cut out to reduce the size of the SDT. In Figure 2(a), the observation nodes are X_4 and X_8, while in Figure 2(b), the observation nodes are X_1, X_4, and X_8. The *Reachable* algorithm selects different sets of reachable nodes, which form different SDTs. More detailed information can be found in Koller and Friedman (2009).

3. Performance-oriented system digital twin

A performance-oriented SDT framework is proposed for the management of infrastructure systems. In this section, the overall design and procedure of the architecture in constructing an SDT are first briefly introduced, then the design philosophies behind its procedure are provided, and finally the detailed steps for its implementation are presented.

3.1. Flowchart and procedure

The performance-oriented SDT framework guides the construction of custom SDTs based on users' inputs including a PM and all data sources. Figure 3 shows the flowchart and procedure of the proposed architecture to customize SDTs. At the root of the performance-oriented SDT is a KG. KG is a powerful technology for capturing and organizing domain-specific information in an interpretable format. This information is expressed as a directed graph, where nodes signify concepts and instances and edges denote the relationships between these nodes (Akroyd et al., 2021; Savage et al., 2022). By delineating these relationships, KGs enhance the accessibility of information, facilitating its interpretation, querying, and updating by computational agents. In Figure 3, KG represents the whole infrastructure system and is the base of all resulting custom SDTs.

Herein, three samples of KGs are provided. The first sample is a KG associated with the risk of bridge networks in Miami-Dade County (Cheng et al., 2024). More details about the KG will be explained in the later section and the case study in this paper is based on this KG. The second sample is a dynamic KG developed by the University of Cambridge (Akroyd et al., 2021). The dynamic KG includes concepts and data describing the real world and is updated in real time by autonomous computational agents. The third sample is developed by the same research group as the second (Savage et al., 2022). It creates new ontologies to extend the dynamic KG to the field of climate and energy supply. These dynamic KGs demonstrate advanced capabilities in data integration for a universal DT. However, they do not incorporate the ability of probabilistic analysis, thus do not align with the purpose of this study.

Given the base graph in Figure 3, the user can provide further inputs to the performance-oriented SDT according

to specific modeling/management objectives. The performance-oriented SDT generates answers to decision support questions under any what-if scenarios such as: given observations $\boldsymbol{O}$, how could an event A affect an event B or a variable C? To answer such questions, the user first identifies observations $\boldsymbol{O}$ and event A (denoted by their corresponding variables) as data nodes. The user can then specify a PM node for event B or designate the variable C as the PM node. Overall, the user needs to identify a PM node and label data nodes as the inputs to the performance-oriented SDT. Then, based on modeling needs/objectives the user can restructure the graph by connecting the PM node to other relevant nodes in the graph.

The following example illustrates the two steps in building a custom SDT. A user wants to establish an SDT to monitor/model/manage the risk associated with the riverine bridges in a bridge network using traffic loads and bridge capacity data. Suppose he/she has an existing KG consisting of 20 bridges (e.g. 12 highway bridges and 8 riverine bridges), with each bridge having multiple nodes such as a node for the risk of the current year, a node for traffic loads each day, and a node for bridge capacity, among others. For his/her risk-management objectives, the user can (1) add the annual risk of riverine bridges as the PM node and label the nodes reflecting traffic loads and bridge capacity as data nodes; (2) connect the PM node to the nodes showing the risk of riverine brides (i.e. 8 risk nodes).

After the KG is restructured and labeled as above, the performance-oriented SDT can generate a custom SDT for the users' objective following three steps:

1. Select a custom trunk: Given the PM node and data nodes, the framework will select a custom trunk, which is a subgraph consisting of relevant nodes including the PM node, data nodes, a set of reachable nodes, and the links between them. This trunk represents the essential parts of the SDT because the nodes within the trunk are the only nodes from the KG that are related to Bayesian updating given the PM and data sources. Then the trunk is cut out from the existing KG. In doing so, the number of selected reachable nodes can be minimized while preserving the ability to approximate and update the PM to desired accuracy given the data nodes. How to select the set of reachable nodes will be discussed later.
2. Construct the BN: The trunk is first expanded to create a new subgraph similar to how branches and twigs grow out of the tree trunk. Specifically, the decision-maker can attach some extra nodes to the trunk based on the modeling need. Although these nodes are not related to Bayesian inference, the objective of adding them is to complement the probabilistic analysis and to improve the modeling accuracy. Given the new subgraph, then a BN can be constructed. Particularly, Monte Carlo simulation is conducted to learn the marginal and conditional probability tables for the root nodes and child nodes of the subgraph, respectively. A BN is then created using the new subgraph and the probability tables.
3. Transform the BN to an SDT: An SDT is implemented using the BN. The SDT builds upon Bayesian inference on the PM node. It needs to automatically determine the query and evidence variables among the BN when performing the inference. Besides, it needs to streamline its modeling of the evolving BN over time. A brief explanation on how to transform the BN into the SDT will be presented later.

3.2. Design philosophies

At the core of the conceptual design of the performance-oriented SDT architecture is storing a directed acyclic graph from which an SDT is created when needed. There are two design philosophies underpinning the architecture of this framework: setting the boundary, and detaching SDTs.

3.2.1. Setting the boundary

Infrastructure systems are large in scale and complex in scope. It is infeasible to build an SDT that encompasses all components, details, and aspects of such infrastructure systems with one single BN. First, it is computationally infeasible to store such a BN because a BN of immense infrastructure systems can take up an extremely large amount of memory space, especially when some child nodes have numerous parent nodes (Byun et al., 2019; Tien & Der Kiureghian, 2016). Second, too many data layers/sources lead to a noisy SDT. Data comes in from all parts of the infrastructure systems at various frequencies (e.g. visual inspections are every two years and structural health monitoring can be continuous). The PM node can be influenced by nodes from close to far away in distance and from spontaneous to frequent occurrences. As a result, the probability distribution of the PM node could be unnecessarily updated with superfluous fluctuations due to irrelevant and noisy data inputs.

In real applications, from the perspective of the stakeholders, the problem they are facing, whether small or large, and the data they can acquire, whether small or big, only involve a portion of the infrastructure systems. Therefore, when choosing nodes for the BN, redundant data sources and irrelevant parts of the infrastructure systems can be selectively discarded. In other words, the boundary should be set for an SDT based on the specific problem the stakeholder wants to analyze and model, and the data that the stakeholder can obtain. Within the performance-oriented SDT, on the one hand, a KG is stored to represent the infrastructure systems taking a small amount of memory space; on the other hand, bounded SDTs are created to overcome the two challenges mentioned above.

3.2.2. Detaching system digital twins

As mentioned previously, various bounded SDTs can be established from the stored KG. Furthermore, the resulting SDTs can work separately to serve different purposes. The

major purpose for detaching these SDTs is for the nodes in the KG to be accessed and used via different independent SDTs. To enable the same node to be used simultaneously by different SDTs, the design philosophy is that SDTs should be detached and work independently. To realize this philosophy, it is proposed that for each new modeling and management task, a new SDT is created from the stored KG.

Consider the following two SDTs. One SDT is derived to monitor the long-term risk of a riverine bridge network, while the other serves as a flood warning system for the same bridge network. For the two SDTs, the nodes of their BNs can be identical because they both monitor the risk of riverine bridges (i.e. same PM) by using the flood discharge data and bridge capacity data (i.e. same data sources). However, updating intervals of the PM are different between the two SDTs. Specifically, the updating interval of the former is on a yearly basis for long-term risk, while the flooding risk in short-term has to be monitored in real time for the latter SDT. Therefore, the processing of flood discharge data should be different (e.g. take the mean values over a year versus hourly, respectively), though the data is from the same sensors. Subsequently, the conditional probability tables are different leading to different BNs and SDTs.

3.3. Implementation details

This section describes the implementation details in three steps to establish an SDT within the performance-oriented SDT framework.

3.3.1. Algorithm for step (1): select a custom trunk

As mentioned previously, the user has defined the PM node and data nodes and restructured the KG. In this section, an algorithm is proposed to select a custom trunk based on the information provided by the user. The algorithm is denoted as $SelTrunk(\boldsymbol{D}, P, \mathcal{G}_r, L_e)$, where $\boldsymbol{D} = \{D_1, D_2, \ldots, D_n\}$ is the set of data nodes, P is the PM node, $\mathcal{G}_r$ is the restructured directed graph, and L_e is the extra length related to the size of the subgraph. L_e is a user input with a default value of 0. The *SelTrunk* algorithm consists of four cuts (see the Supplementary Appendix). The algorithms for the first to fourth cuts are shown in Algorithms 1 to 4, respectively, while the complete *SelTrunk* algorithm is shown in Algorithm 5, where the four cuts are integrated sequentially. The general principle for creating a custom trunk is to remove all the nodes that do not affect/update the PM node, thus effectively controlling the size of the directed acyclic graph and fundamentally improving the computational efficiency of the SDT. Detailed explanations for the four cuts are as follows:

1. $FirstCut(\mathcal{G}_d, \boldsymbol{D}, P, L_e)$. The first cut significantly trims the size of the BN in constructing an SDT. It removes nodes that are far away from the PM node. Specifically, the cutoff distance is the farthest distance of the data nodes. The intuitions to design the algorithm in this manner are that (1) the farther away a node is from the PM node, the less influence it will have on the PM node and (2) the influence of a node will be blocked if there exists a data node on its path to the PM node.
2. $SecondCut(\mathcal{G}_d, \boldsymbol{D}, P)$. The second cut discards nodes that are independent of the PM node given the data nodes. This is done using the *Reachable* algorithm.
3. $ThirdCut(\mathcal{G}_d, \boldsymbol{D}, P)$. The third cut prunes the branches that have no PM node and data nodes. Figure 4 illustrates the *ThirdCut* algorithm. Specifically, Figure 4(a) is the remaining graph after removing X_6 and X_{12} of Figure 2(a). The *ThirdCut* algorithm then identifies and cuts the two branches in Figure 4(a) that can be pruned, leading to what is shown in Figure 4(b) after pruning. The design philosophies underlying this algorithm are that if a branch does not have a data node, then it will not provide new information that can update the PM node. Therefore, this whole branch is irrelevant to the SDT in construction, though it can contribute to the probabilistic analysis.
4. $FourthCut(\mathcal{G}_d, P)$. The fourth cut is a postprocessing step. It cleans up the isolated nodes in the graph that have no pathway connecting to the PM node.

3.3.2. Implementation details for step (2): construct the Bayesian network

The custom trunk is important for Bayesian updating. However, it may not be enough to support probabilistic analysis of desired accuracy. Therefore, extra nodes are attached to the trunk in a way similar to how the branches and twigs grow out of a tree trunk. For each of the trunk nodes, its parent nodes and child nodes in the KG are identified. If a parent node can improve the probabilistic modeling of the custom trunk from the perspective of the decision-maker, that parent node will be attached to the trunk. Similarly, if a child node of the trunk node serves as a data source, it will also be attached to the trunk.

For example, from Figure 4(a,b), the right branch stemming from X_2 is retained while the left branch of P_1 including X_1, X_3, X_{10}, and X_{11} is removed. However, complete removal of the left branch can impede the accuracy in modeling uncertainty associated with P_1. On the one hand, it is better to incorporate the left branch into the modeling, on the other hand, the whole left branch is superfluous. Hence, if the decision-maker wants to improve the probabilistic modeling, it is suggested to attach X_1 to P_1 and use the data for X_1. Otherwise, a default and deterministic value needs to be used for X_1. The above procedure needs to be carried out for all the nodes in the trunk.

After creating the new subgraph, the next step is to elicit the marginal and conditional probability tables for constructing the BN. This involves discretizing and sampling the nodes (i.e. random variables) and estimating the probability distributions. Consider the conditional probability tables for X_1, X_2, and X_4 in Figure 1 as an example. First, X_1, X_2, and X_4 are discretized. Let $X_{1,i}$ be the ith state of X_1 that represents the values in the ith interval of X_1, $(X^L_{1,i}, X^U_{1,i}]$, where $X^L_{1,i}$ and $X^U_{1,i}$ are the lower and upper bounds, respectively. Consequently, $X_1 = X_{1,i}$ means that X_1

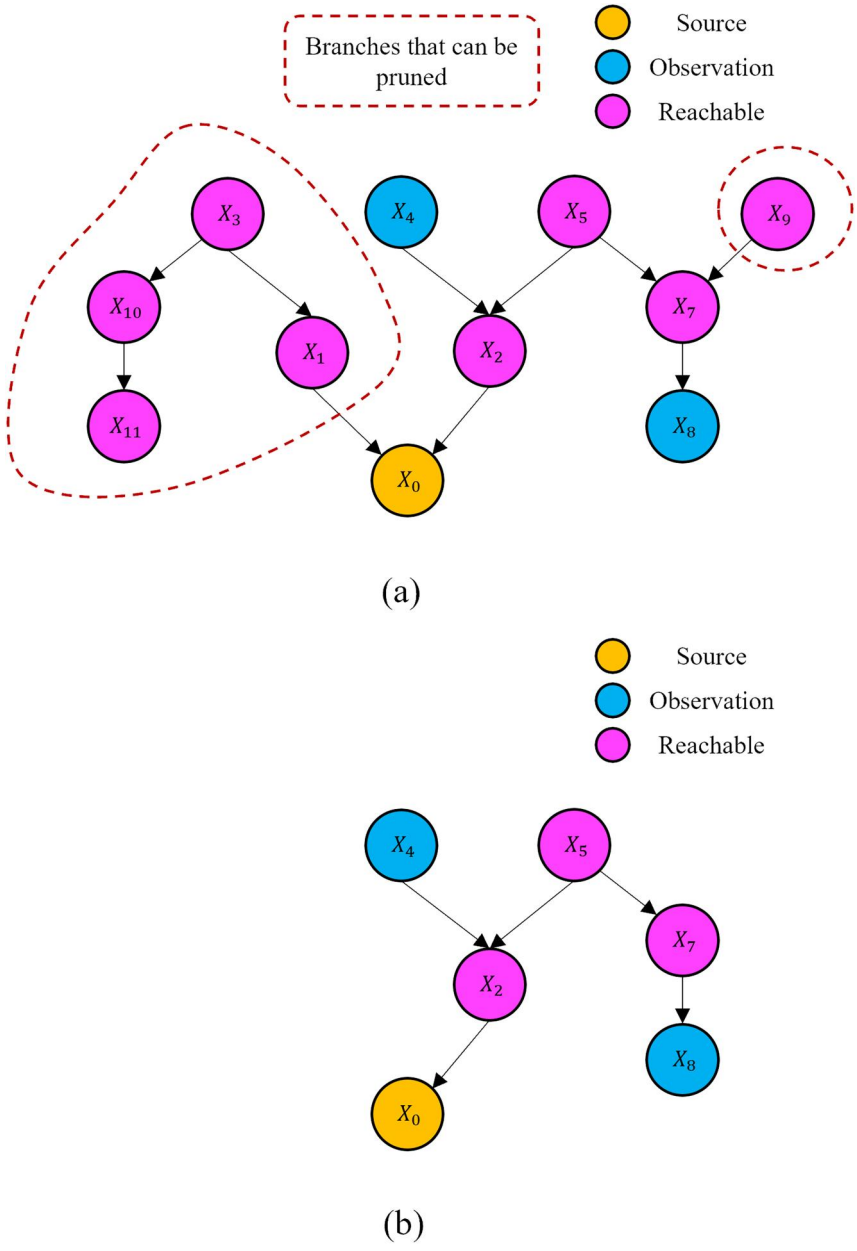


Figure 4. Illustration of the *ThirdCut* algorithm: (a) branches that are identified by the algorithm, (b) the subgraph after cutting the identified branches.

falls into the interval $(X^L_{1,i}, X^U_{1,i}]$. Notations for X_2 and X_4 are similar. Monte Carlo simulations are used to uniformly sample X_1 and X_2. For each sample, X_4 is determined through expert knowledge or computation. A probability value in the conditional probability table is then computed as:

$$P\left(X_4 = X_{4,k} | X_1 = X_{1,i}, X_2 = X_{2,j}\right) = \frac{N(X_1 = X_{1,i}, X_2 = X_{2,j}, X_4 = X_{4,k})}{N(X_1 = X_{1,i}, X_2 = X_{2,j})} \quad (2)$$

where $N(X_1 = X_{1,i}, X_2 = X_{2,j})$ is the number of samples that satisfy $X_1 = X_{1,i}$ and $X_2 = X_{2,j}$, and $N(X_1 = X_{1,i}, X_2 = X_{2,j}, X_4 = X_{4,k})$ is defined in the same way. If X_1, X_2, and X_4 have N_1, N_2, and N_4 states, respectively, then $N_1N_2N_4$ probability values need to be computed to determine the conditional probability table. More detailed information on constructing BN can be referred to Bensi (2010).

3.3.3. Implementation details for step (3): transform the Bayesian network to a system digital twin

Essentially, the objective of an SDT is to perform Bayesian inference on the PM node over time. Therefore, after constructing the BN, Step (3) is to utilize the BN to streamline and conduct Bayesian inference. This section briefly introduces classification of the BN nodes and then discusses how to perform the inference based on the classification. Please note that implementation details and an illustrative example (i.e. an SDT to monitor life-cycle risk of a bridge network) for this Step (3) can be found in earlier work of the authors (Cheng & Gao, 2023).

BN nodes are classified based on their functions and temporal properties. Based on the functions, BN nodes are classified into three types: PM nodes, data nodes, and analysis nodes. The definitions of PM node and data nodes have been mentioned in previous sections, all the remaining BN nodes are analysis nodes. A data node is further categorized as a continuous data node (CDN) if its data reporting frequency is higher than the updating frequency of the PM node. Otherwise, it's an intermittent data node (IDN). Based on the temporal property, the nodes are classified as time-dependent or otherwise. When PM nodes, IDN, and analysis nodes are time-dependent, they are denoted as PM-T, IDN-T, and AN-T, respectively, otherwise, they are denoted as PM-N, IDN-N, and AN-N, respectively.

The frequency of Bayesian inference is the same as that of the PM node. At each inference, the query variables are PM-T, PM-N, IDN-T with no data, and AN-T, while evidence variables are IDN-T with data, IDN-N with data, and CDN. To represent the previous time step of a time-dependent node, a parent node is added to it. Between two inferences, probability distributions of time-dependent nodes (i.e. AN-T, IDN-T, and PM-T) are copied to their parent nodes. In summary, the node classification can streamline Bayesian inference by enabling data nodes to receive real-world updates at specific intervals, which are then used to perform Bayesian updating on performance nodes.

4. Illustrative example 1: a hypothetical graph

A hypothetical graph with 36 nodes is used to demonstrate the construction of a performance-oriented SDT. This graph, as shown in Figure 5, is considered the KG representing the infrastructure systems and is stored in the performance-oriented SDT. The KG has two components, which can represent two different analyses of the same physical asset (e.g. failure consequence evaluation and reliability analysis of a structure), two physical assets (e.g. two roads to connect the same origin and destination), or two infrastructure systems (e.g. power and water distribution systems). The analyses of two components are independent, but they have the potential to be combined to perform other probabilistic analyses.

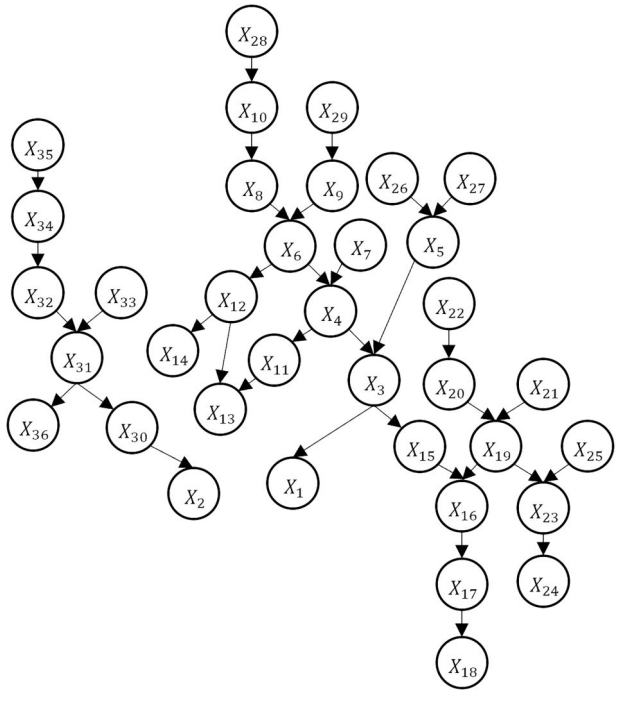


Figure 5. A hypothetical knowledge graph with two components.

4.1. Select a custom trunk

Three scenarios are used to demonstrate how to select trunks. Suppose the decision-maker attempts to create three SDTs consisting of PM nodes and data nodes for the following scenarios. In scenario 1, the two components in Figure 5 are connected by adding the PM node P_1 as the child node of X_1 and X_2, while the data nodes are X_5 and X_{11}. Scenario 2 uses the same PM node as in Scenario 1, while data nodes are X_5, X_{11}, X_{17}, and X_{19}. For scenario 3, the PM node P_1 is the child node of X_{18} and X_{24}, while data nodes are X_5, X_{11}, X_{17}, and X_{19}. For all three scenarios, the extra length is 0.

Before running the *SelTrunk* algorithm, the graph needs to be restructured based on the decision-maker's inputs. Figures 6(a), 7(a), and 8(a) show the restructured graphs (i.e. including the green, blue, orange, and white nodes) for scenarios 1, 2, and 3, respectively. These three figures also illustrate the *FirstCut* algorithm. The blue nodes and the orange node in the restructured graph indicate the data nodes and PM node, respectively, which are provided by the users. Green nodes are selected during the first cut because their distances to the PM node are smaller than a computed threshold, while white nodes are discarded in the first cut because they are too far away to have significant impact on the PM node. Since the KG can be very large, the first cut significantly reduces the size of the custom trunk.

All the nodes in Figures 6(b), 7(b), and 8(b) show the resulting graph after the first cut. The green nodes are selected by the *SecondCut* algorithm. This algorithm, based on the *Reachable* algorithm, can eliminate nodes that do not have influence on the PM node. The *SecondCut* also identifies redundant data nodes in cases where the decision-makers over-supply too many repeating data pieces. As shown in Figure 8(b), the influence of data nodes X_5 and X_{11} is blocked by data nodes X_{17} and X_{19}. Therefore, for scenario 3, X_5 and X_{11} are redundant data sources given the presence of X_{17} and X_{19}. On the contrary, scenario 2 has the same data nodes as scenario 3 has. However, X_5, X_{11}, X_{17}, and X_{19} are all statistically correlated with the PM node, hence, they are all useful data sources. The implication of the results is that the decision-maker can provide as many data sources as they think are useful for the SDT and let the performance-oriented SDT algorithm figure out what to be used in building the custom SDT to address his/her specific project needs.

The graphs resulting from the second cut are shown in Figures 6(c), 7(c), and 8(c). Though all the nodes in the figures can have influence on the PM nodes, they may be irrelevant from the perspective of DTs. For example, in

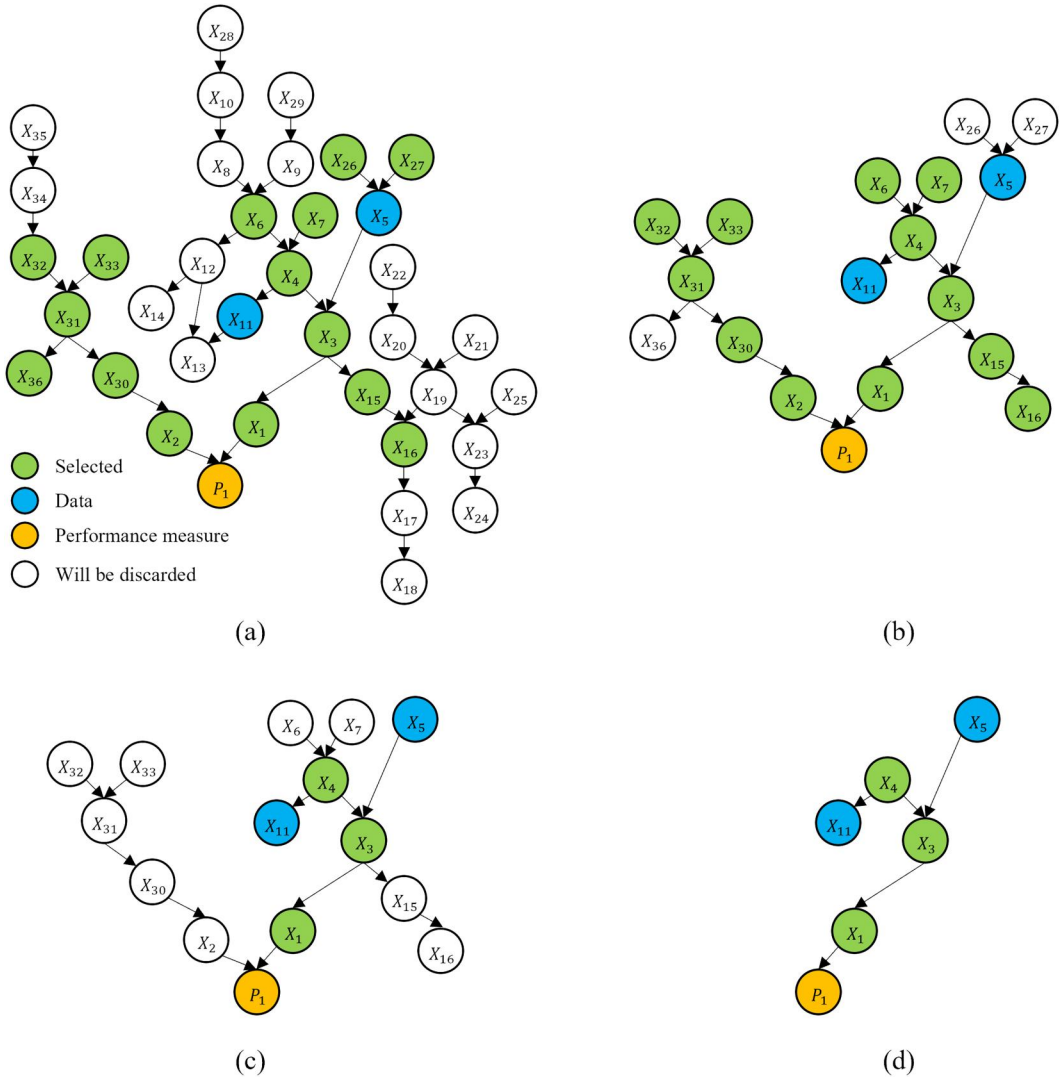


Figure 6. Illustration of the *SelTrunk* algorithm for scenario 1: (a) first cut, (b) second cut, (c) third cut, and (d) fourth cut.

Figure 6(c), the branch including X_2, X_{30}, X_{31}, X_{32}, and X_{33} has no data nodes. Hence, this branch cannot trigger Bayesian inference on the PM node. As a result, this branch is omitted by the *ThirdCut* algorithm. In Figure 6(c), X_{15} and X_{16} are not selected due to the same reason. In contrast, in Figure 7(c), the branch of X_{15}, X_{16}, X_{17}, and X_{19} is selected because X_{17} and X_{19} are data nodes. In this case, X_{15} and X_{16} should be kept since they act as a bridge to transfer the influence of X_{17} and X_{19} to P_1.

Please note that the *ThirdCut* algorithm will also remove certain single parent or child nodes of the nodes in the remaining trunk. For example, in Figure 6(c), X_4 is selected into the trunk, but its parent nodes X_6 and X_7 are removed. For another example, in Figure 8(c), X_{23} is selected into the trunk, while one of its parent nodes X_{25} is removed. Discarding these branches and nodes such as X_2, X_{30}, X_{31}, X_{32}, and X_{33} in Figure 6(c) and X_{25} in Figure 8(c) does not imply that they are redundant. Rather, it only means that they don't provide useful information for PM node inference in the constructed SDT. If the decision-maker deems them important for probabilistic analysis, a branch can be represented by a special node, as to be demonstrated in next section.

Figures 6(d), 7(d), and 8(d) show the graphs after the third cut and demonstrate the *FourthCut* algorithm. The algorithm is a post-processing step to clean isolated nodes. As shown in Figures 6(d) and 7(d) for scenarios 1 and 2, no nodes are removed in this step since there are no isolated nodes. For scenario 3, Figure 8(d) shows that the isolated nodes X_5 and X_{11} are discarded. Overall, Figures 6–8 illustrate how to select a custom trunk based on the users' inputs. The *SelTrunk* algorithm defines and implements a versatile as well as rigorous architecture for building performance-oriented SDTs that enable event-responsive reasonings and support decision making with flexibility to accommodate various use cases.

4.2. Construct the Bayesian network

Scenario 1 is used to illustrate how to construct a BN. Figure 6(d) shows the graph after four cuts. However, this graph may not be enough to perform a probabilistic analysis

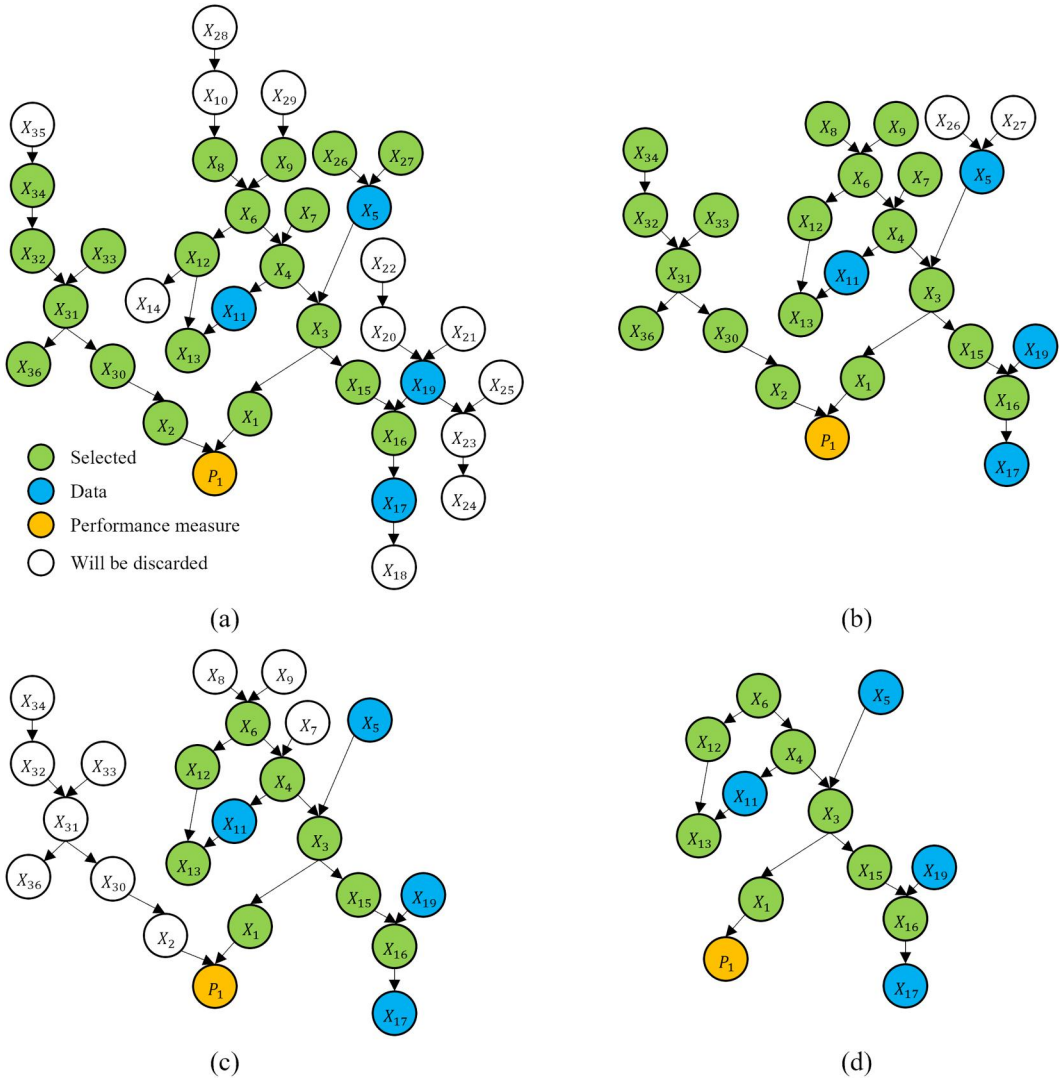


Figure 7. Illustration of the *SelTrunk* algorithm for scenario 2: (a) first cut, (b) second cut, (c) third cut, and (d) fourth cut.

at a desired level of accuracy. For this, the user can attach necessary nodes to the trunk. For example, the restructured KG in Figure 6(a) shows that the PM node is affected by two parent nodes, X_1 and X_2, while in previous steps the trunk included X_1 and removed the branch originated from X_2. On the one hand, chances are that including this branch can significantly improve the modeling accuracy. On the other hand, the whole branch is not necessary for SDT. Therefore, as a compromise, the user can attach X_2 to P_1 and use or collect data to characterize X_2. X_2 is considered as the special node to represent the whole branch. It is further assumed that X_2 is time-dependent. Based on these assumptions, a parent node $X_{2,0}$ is attached to X_2. Figure 9 shows the final custom trunk for the SDT of scenario 1.

For this illustrative example, a data-driven system reliability analysis is performed with the assumptions of the following relations:

$$P_1 = \begin{cases} 0 & \text{if } X_1 + X_2 - 0.25\sqrt{X_1 X_2} \leq 6.5 \\ 1 & \text{if } X_1 + X_2 - 0.25\sqrt{X_1 X_2} > 6.5 \end{cases} \quad (3)$$

$$X_1 = 7 - 0.05X_3^2 - 0.1X_3 + \epsilon_1 \quad (4)$$

$$X_3 = 2X_4 + X_5 - 0.25X_4 X_5 + \epsilon_2 \quad (5)$$

where ϵ_1 and ϵ_2 are modeling errors in normal distribution with a mean of 0 and a standard deviation of 0.25. X_1, X_2, X_3, X_4, and X_5 are all continuous random variables and are discretized. For the discretization, the state values of all the random variables represent an interval of plus and minus 0.5. For example, $X_4 = 1$ represents that X_4 falls in the interval of $(0.5, 1.5]$. The states of X_4 are from 1 to 5 and the states of X_5 are from 3 to 8, while based on the computation results, the states of X_1 and X_3 are from 1 to 6 and from 3 to 10, respectively.

It is also assumed that X_2 follows a Markovian process. The states of X_2 and $X_{2,0}$ are from 1 to 5 with 5 and 1 indicating the best and worst performance, respectively. Each year, X_2 has a probability of 0.95 to stay in its previous state and a probability of 0.05 to move to a lower state. If X_2 reaches state 1 (i.e. the worst state), it will stay in state 1. The transition probability from $X_{2,0}$ to X_2 is:



Figure 8. Illustration of the *SelTrunk* algorithm for scenario 3: (a) first cut, (b) second cut, (c) third cut, and (d) fourth cut.

$$\boldsymbol{P}_{X_2|X_{2,0}} = \begin{bmatrix} 0.95 & 0.05 & 0 & 0 & 0 \\ 0 & 0.95 & 0.05 & 0 & 0 \\ 0 & 0 & 0.95 & 0.05 & 0 \\ 0 & 0 & 0 & 0.95 & 0.05 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{6}$$

Consider a system reliability analysis with two components, X_1 and X_2. The component represented by X_2 has deterioration, which is modeled by the Markov chain. For the analysis of other components, X_4 and X_5 can be considered as load effects. With larger loads, its performance represented by X_1 is worse. X_{11} is considered as the measurement of X_4. Measurement distribution is assumed as: If X_4 is in states 2, 3, or 4, X_{11} has a probability of 0.6 to be in the same state, and probabilities of 0.2 for both plus and minus 1; If X_4 is in state 1, X_{11} has a probability of 0.8 to be the same, and a probability of 0.2 to be state 2; If X_4 is in state 5, X_{11} has a probability of 0.8 to be the same, and a probability of 0.2 to be state 4. The measurement matrix is:

$$\boldsymbol{P}_{X_{11}|X_4} = \begin{bmatrix} 0.8 & 0.2 & 0 & 0 & 0 \\ 0.2 & 0.6 & 0.2 & 0 & 0 \\ 0 & 0.2 & 0.6 & 0.2 & 0 \\ 0 & 0 & 0.2 & 0.6 & 0.2 \\ 0 & 0 & 0 & 0.2 & 0.8 \end{bmatrix} \tag{7}$$

With the relations between random variables, data for all the random variables is simulated to obtain conditional probability tables. Values are also assigned to root nodes in the BN, which include $X_{2,0}$, X_4, and X_5. Specifically, $X_{2,0}$ takes on a value starting from 5, which has the state probability of $[1, 0, 0, 0, 0]$. X_4 is assigned a triangular distribution $[0.08, 0.24, 0.36, 0.24, 0.08]$, and X_5 also a triangular distribution $[1/18, 1/6, 5/18, 5/18, 1/6, 1/18]$. With all the marginal probability tables and conditional probability tables established, the BN is constructed and can be used for SDT inference.

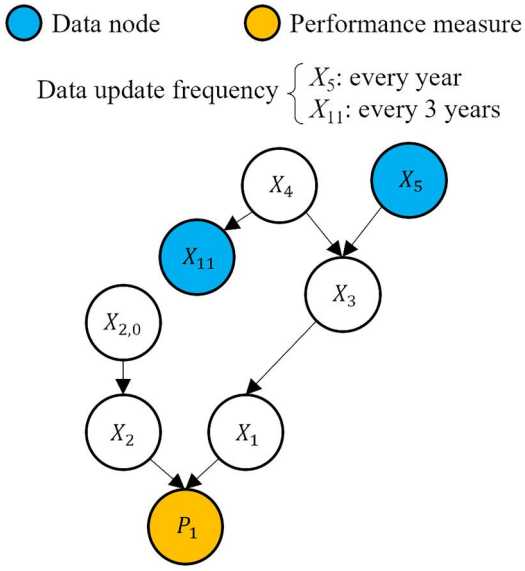


Figure 9. Custom trunk for the SDT of scenario 1 (Figure 6).

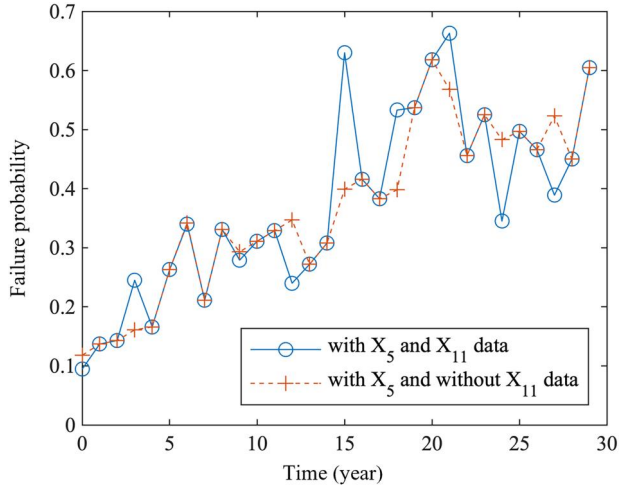


Figure 10. Failure probability profile.

4.3. Transform the Bayesian network to a system digital twin

This example for data-driven reliability analysis uses two data sources, X_5 and X_{11}. To illustrate how the SDT works in this context, it is further assumed that X_5 and X_{11} are a CDN and an IDN-N, respectively. Data for them is generated through Monte Carlo simulation and then processed through the SDT to perform Bayesian inference. Specifically, data from X_5 follows a triangular distribution with a lower limit of 2.5, an upper limit of 8.5, and a mode of 5.5, while data from X_{11} follows a triangular distribution with a lower limit of 0.5, an upper limit of 5.5, and a mode of 3. Please note that after discretization, data of X_5 and X_{11} have the assigned marginal probability distributions of X_5 and X_{11} in the BN, respectively. Moreover, it is assumed that X_{11} is observed every three years with the first observation at year 0. This numerical example spans from year 0 to year 29.

As mentioned previously, X_5 is a CDN and X_{11} is an IDN-N. P_1 is further classified as a PM-N and X_2 as an AN-T, while all the other nodes are classified as AN-N. Based on the classification of the nodes, the SDT can automatically determine the evidence and query variables in each update. Figure 9 shows the evidence variables and their updating frequencies for illustrative purposes. P_1 is the PM node to be monitored, so it is always a query variable. X_2 is also a query variable, because it is time-dependent. Between the consecutive Bayesian inferences, the probability distribution of X_2, which is the query result, is copied to $X_{2,0}$. Data from X_5 is always available, so it is an evidence variable all the time. X_{11} is only available every three years, so X_{11} is an evidence variable at years 0, 3, 6, and 9.

The objective of this reliability analysis is to monitor the failure probability (i.e. the probability of $P_1 = 0$). The failure probability profile is shown in Figure 10. Though with fluctuation, the underlying trend is that the probability of failure increases over time. This is due to the deterioration of X_2. To demonstrate the ability of SDT to incorporate newly available data in decision analysis, Figure 10 compares the failure probability profiles with and without data from X_{11}. The results show that when receiving data only from X_5, such as at years 1, 2, 4, and 5, the predicted failure probabilities of two cases are the same. On the contrary, when data is available from both X_5 and X_{11}, the failure probabilities of the two cases are different, as evidenced by the results at years 0 and 3.

Moreover, when the observed value of X_{11} is large, the failure probability of the case with X_{11} data is larger than that of the case without X_{11} data. For example, at years 3 and 15, the observed X_{11} is 4 and 5, respectively, which are larger than the mode. This generally leads to larger failure probabilities than those without observation. On the contrary, when the observed X_{11} is smaller than the mode, it will generally lead to a smaller failure probability. Together, these results demonstrate that the SDT can leverage information from relevant data from different sources to perform Bayesian inference on the PM node.

For demonstration, the trunk of scenario 1 is selected to be built into an SDT. As previously shown in Figure 3, within the performance-oriented SDT, for each set of the PM and data sources that the user specifies, a custom SDT can be built. In this illustrative example, three scenarios are presented with each scenario having a set of PM nodes and data sources. Figure 11 shows that for each scenario, a trunk is selected using the *SelTrunk* algorithm and is then transformed into an SDT. Figure 11 also shows that as the user provides more sets of inputs, the performance-oriented SDT framework can generate more custom trunks giving birth to corresponding SDTs that function independently.

5. Illustrative example 2: a bridge network application

Another real-world scenario on monitoring the risk associated with bridges is used to illustrate how the proposed framework can be used to solve practical problems. First, an ontology is created to represent the risk of bridge network

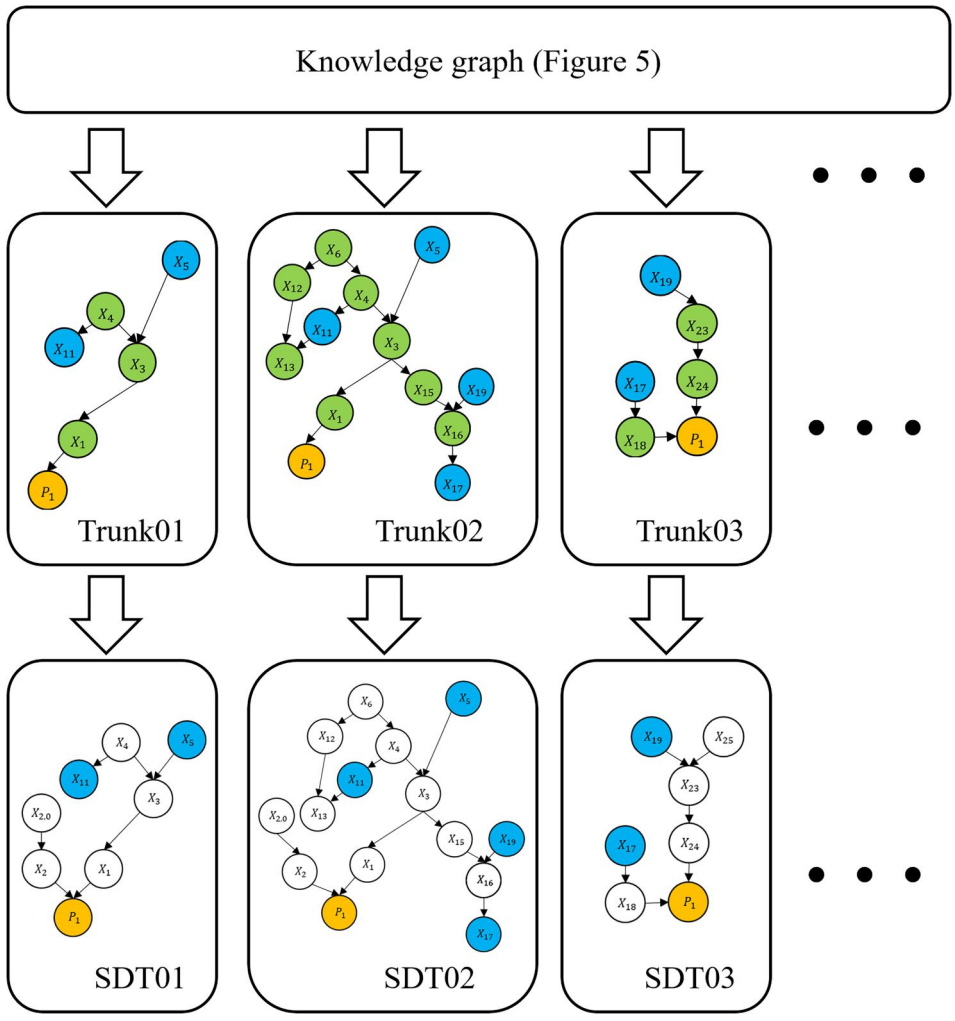


Figure 11. Performance-oriented SDT with independent SDTs.

and risk-related data sources in Miami-Dade County. Data sources to update the risk include the superstructure and substructure condition ratings of bridges, flow and stage of river gages, and annual average daily traffic (AADT) of portable traffic monitoring sites. The ontology defines dependencies between concepts across these three interlinked systems: bridge, traffic, and river systems. Figure 12 shows how these cross-system data are utilized to determine the bridge risk, where concepts from different systems are marked in various colors.

Specifically, the annual risk of a bridge is the product of the failure probability and failure consequence of the bridge (Cheng & Frangopol, 2023; Liu et al., 2020), therefore the ontology links the failure probability and failure consequence of a bridge to the failure risk. The superstructure and substructure condition ratings represent the capacity of a bridge and thus are related to the failure probability of a bridge (Akgül & Frangopol, 2004), while the flow and stage of river gages represent the flood loads on the riverine bridges and thus are also related to the failure probability (Liu et al., 2020). Statistical correlations exist between the flow and stage of individual river gages and across river gages. This is modeled by connecting the correlated nodes through a common source node. One of the consequences of bridge failure is the detour of vehicles, therefore, AADT of nearby monitoring sites is used to determine the failure consequence (Liu et al., 2020).

Following the ontology, a large-scale KG is generated to represent the risk management of all bridges in Miami-Dade County. The KG is structured with nodes representing random variables involved in the calculation of network risk and edges denoting the dependencies between these random variables. The dependencies can be interpreted in two primary ways: (a) Causal relationships: One variable serves as the cause and another represents the effect. For example, the substructure condition rating, superstructure condition rating, along with river flow and stage, are used to estimate the failure probability of bridges. (b) Common source relationships: Variables exhibit correlations due to shared upstream influences or latent common causes. Examples include flow and stage data collected from different river gages but originating from the same river. Overall, in this KG, there are 1,034 bridges, 1,496 portable monitoring sites, and 8 river gages. More detailed information about generating a large-scale KG can be referred to Cheng et al. (2024).

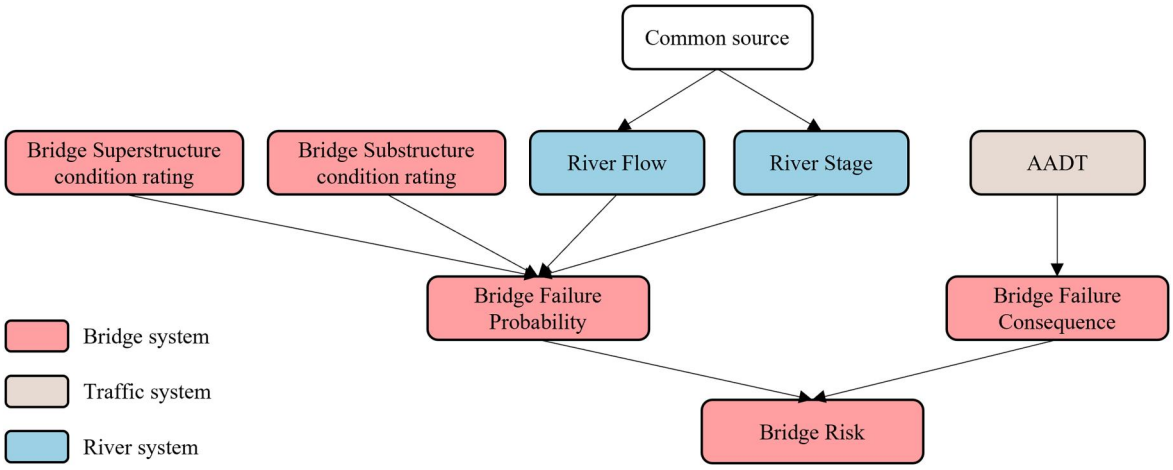


Figure 12. Ontology for determination of bridge risk using condition ratings, AADT, and streamflow (adapted from Cheng et al. (2024)).

Suppose the user wants to monitor the annual risk of the bridge network and the available data sources are 8 river gages. Following the performance-oriented SDT framework, the KG needs to be restructured. Specifically, a node 'Network Risk' is added as the PM node and all the nodes that belong to the 'Bridge Risk' are connected to the PM node. The 8 river flow nodes and 8 river stage nodes are labeled as data nodes. With the user's inputs, the graph is restructured, shown in Figure 13(a). In total, there are 6,687 nodes and 15,638 edges. The number of nodes of different types in the restructured graph is summarized in Table 1. In Figure 13(b), a subgraph of the restructured graph is selected to illustrate all the nodes related to the bridge with the ID 870013. This subgraph corresponds to the ontology in Figure 12 except that a network risk is added. For Bridge 870013, there exists a list of bridge-related nodes including risk, failure probability, failure consequence, superstructure condition rating, substructure condition rating. There are also river gage-related and AADT nodes associated with this bridge, with the ID shown in Figure 13(b).

KG enables the integration of dependencies across various spatial scales within a system (e.g. linking structural components to individual bridges and subsequently to the bridge network) and across interconnected systems (e.g. capturing interdependencies between bridges, traffic monitoring facilities, and river gages). Specifically, for each bridge, the condition ratings of its substructure and superstructure are linked to its failure probability. The individual bridges are further integrated into the bridge network by embedding the geospatial information of infrastructure components. The connectivity between nodes in the KG is dictated by spatial proximity. River nodes can be connected to the bridge nodes only if the river intersects the bridge and the distance between the river gage and the bridge is within 5 km. Similarly, for the AADT nodes, the connections are built between each bridge and its 10 nearest traffic monitoring sites along the corresponding roadway. Figure 14 shows the geographic locations of bridges, traffic monitoring facilities, and river gages in Miami-Dade County. Three representative bridges are zoomed in to demonstrate how bridges are connected to traffic monitoring facilities and rivers. These geographic connections are consistent with the linkages in KG, ensuring that the KG captures dependencies across the three systems.

However, the large-scale restructured graph will present a significant computational challenge when it transforms to BN. For example, the BN occupies up to 73,362.9 MB of memory to store the conditional probability tables when the number of node discrete states is 4. The performance-oriented SDT framework is proposed to overcome these limitations. Using the *SelTrunk* algorithm, the selected custom trunk has 71 nodes and 166 edges, as shown in Figure 15. The number of nodes of different types in the custom trunk is summarized in Table 1. The BN transformed from the custom trunk occupies 625.3 MB of memory to store the conditional probability tables. Thus, both the graph size and memory usage of the custom trunk are approximately 1% of those of the restructured graph. This demonstrates that the proposed methodology can significantly improve the computational and storage efficiency of SDTs.

The custom trunk is transformed into a BN to perform risk reference. The construction of BN involves learning marginal and condition probability tables from the dataset. Herein, the dataset is generated using Monte Carlo simulation (Bensi, 2010). Specifically, the data ranges for river flow and stage are first determined from the measured data. The samples of river flow and stage are then drawn uniformly from their respective ranges to calculate the failure probability, failure risk, and network risk. The data from all nodes need to be discretized before being used to learn the conditional probability tables. The discretization of the river flow and stage is done by dividing the range of the data into 8 equal-width intervals. The failure probability, failure risk, and network risk are divided into discrete intervals based on percentiles, where the 20th, 40th, 60th, 70th, 80th, 90th, 95th percentiles are used. This discretization allows higher risks to gain more attention.

A future decision analysis scenario is hypothesized to demonstrate how SDT performs the risk inference. River flow and stage are determined as data nodes that are updated every year. These data are generated using a growth coefficient with a mean of 1% and a variance of 1%. In

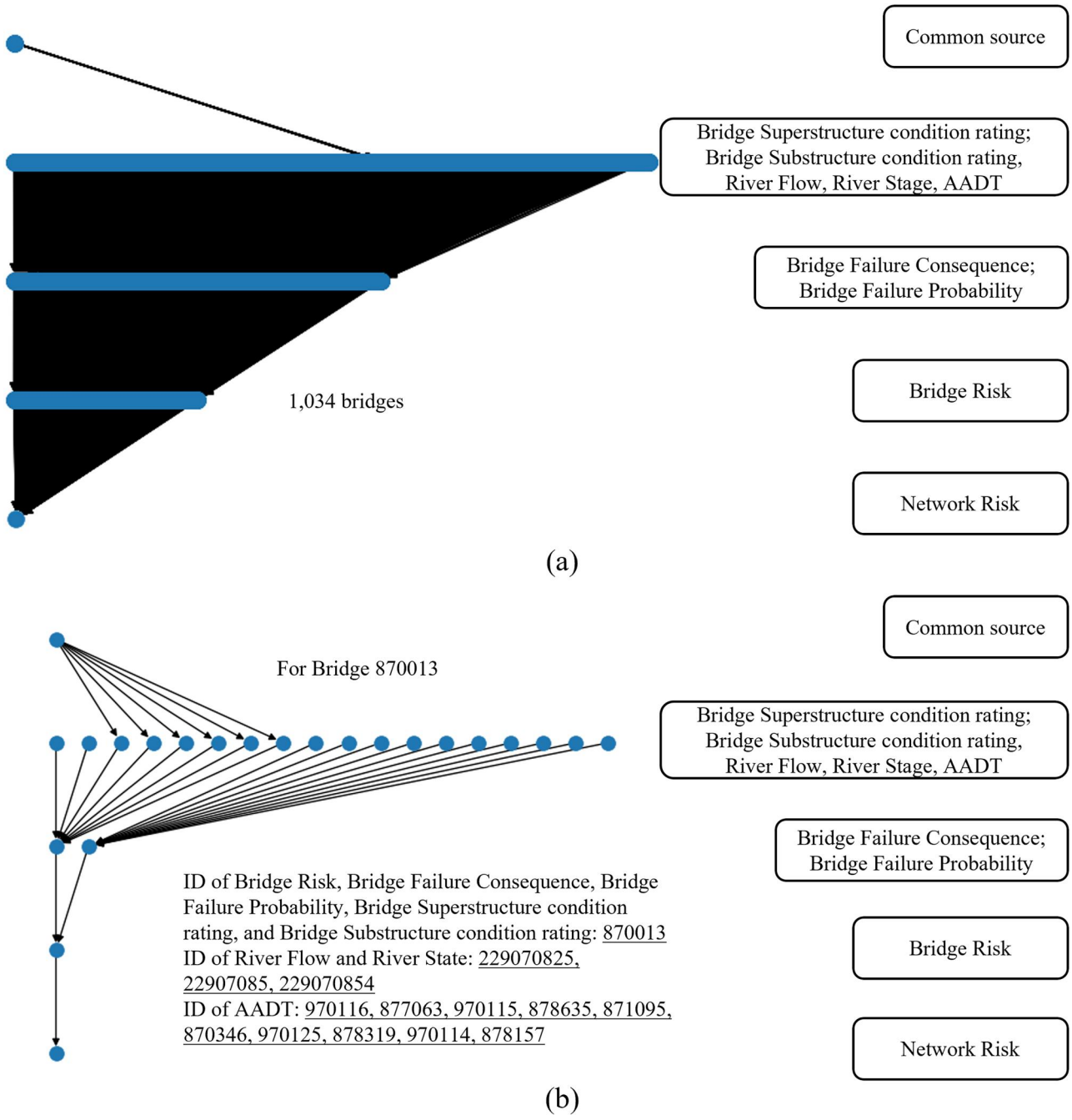


Figure 13. (a) Restructured graph for Miami-Dade county bridge network (6,687 nodes and 15,638 edges) and (b) illustrative graph for bridge 870013. Note: node types on the right.

Table 1. Comparison of number of nodes across different types in restructured graph and custom trunk.

Node type	Number of nodes: restructured graph	Number of nodes: custom trunk
Network risk	1	1
Bridge risk	1,034	27
Bridge failure consequence	1,034	0
Bridge failure probability	1,034	27
Bridge superstructure condition rating	1,034	0
Bridge substructure condition rating	1,034	0
River flow	8	8
River stage	8	8
AADT	1,496	0
Common source	4	0
Total	6,687	71

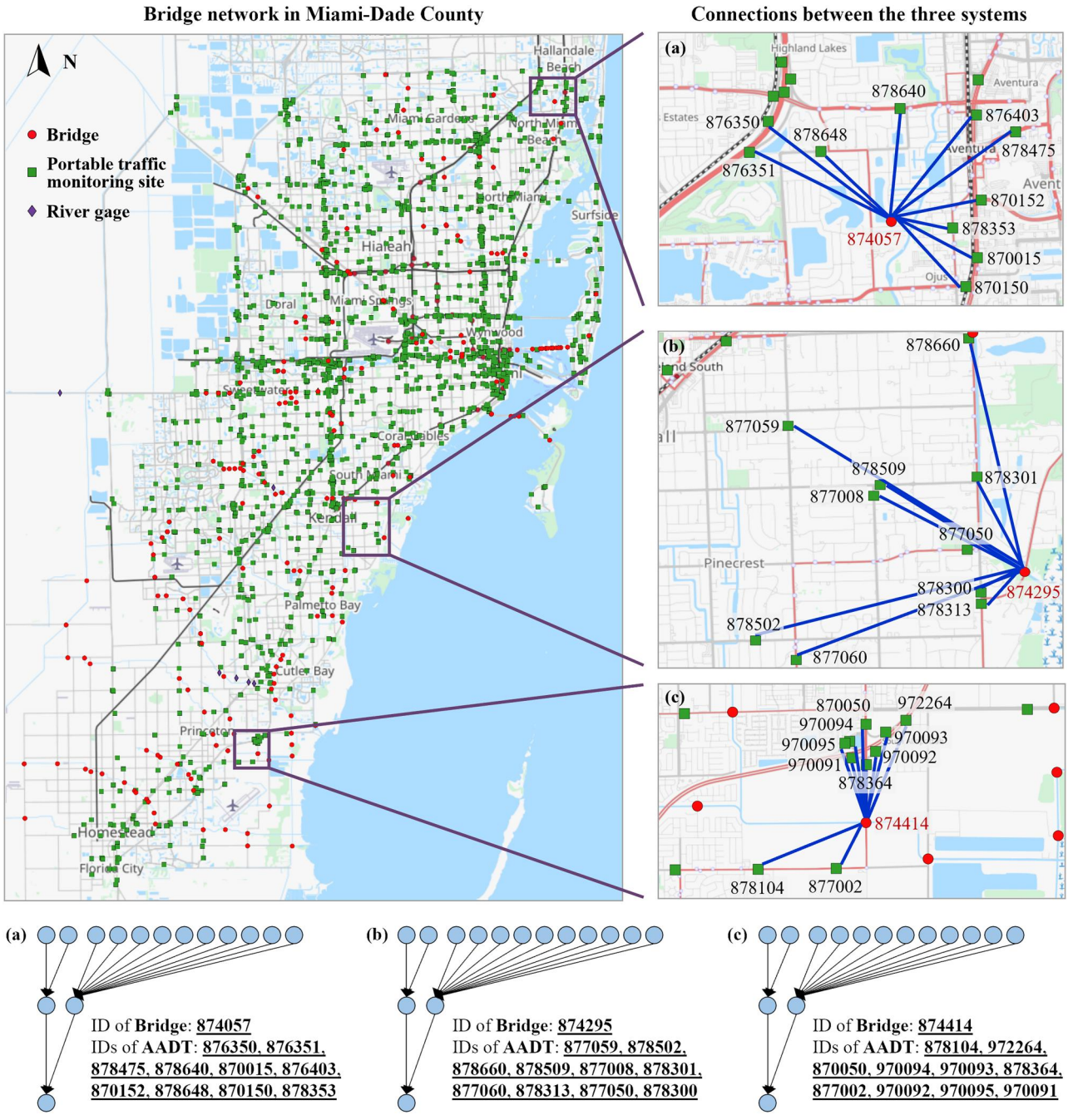


Figure 14. Integration of the topology of three systems into the KG.

other words, the subsequent year's data is derived by multiplying the previous year's data by the growth coefficient. The inferred risk profile of the bridge network is shown in Figure 16. Generally, the risk of the bridge network exhibits a consistent trend with river flow and stage data, gradually increasing over the years, as shown in Figure 17(a). However, in years 3 and 9, a decline in risk is observed. This can be attributed to the lower river flow and stage in these years, as illustrated in Figure 17(b).

The above results show the capacity of the proposed SDT in updating the risk of the bridge network using available observations. This process exemplifies the concept of 'physical-to-virtual twinning', where physical measurements are input into the digital space to update the state of virtual representation. Once the SDT is updated, it can identify bridges with high risk levels and mark them as vulnerable, allowing for prioritized predictive maintenance. This step demonstrates the concept of 'virtual-to-physical twinning', where insights derived from the digital entity inform real-world decision-making and thus optimize the management of infrastructure assets.

6. Limitations and future work

The current framework, while demonstrating the feasibility of implementing DTs for complex systems using custom

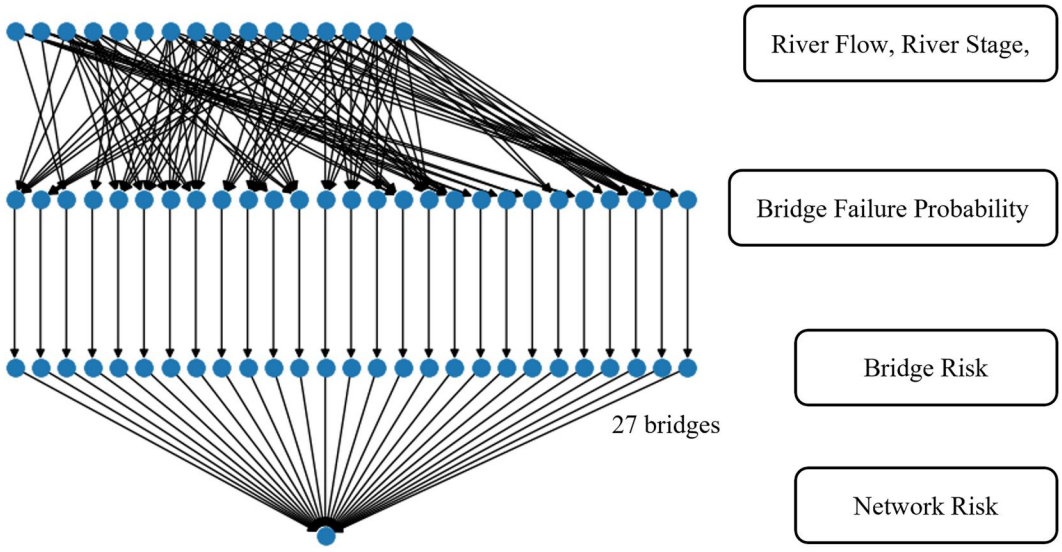


Figure 15. Custom trunk with node types on the right (71 nodes and 166 edges).

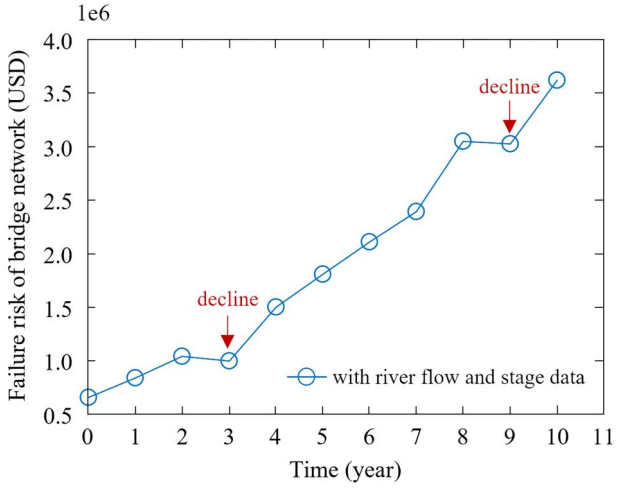


Figure 16. Failure risk profile of bridge network in Miami-Dade county.

BNs, has certain limitations. First, the performance-oriented SDT is built on the premise that the KGs for complex systems at the city or county scale are available. However, such large-scale KGs are limited in practice. Future work should be devoted to developing methodologies for automated KG generation tailored to complex infrastructure systems. Second, while the current framework integrates the spatial geometry of the city/county in the KG, it lacks integration of component-level geometry at a finer resolution. Future work should address this gap by incorporating BIM standards to enable detailed geometry representation of individual assets, thereby enhancing the fidelity of the digital entities. Lastly, the current framework primarily concentrates on the computational aspects of DTs at the system level, without incorporating other modeling enabling technologies within the DT architecture such as sensors, simulations, data-driven models, model predictive control, among others (Broo et al., 2022, Thelen et al., 2022). Future efforts should be directed toward integrating these DT modeling enabling technologies with the proposed framework to develop a comprehensive DT platform.

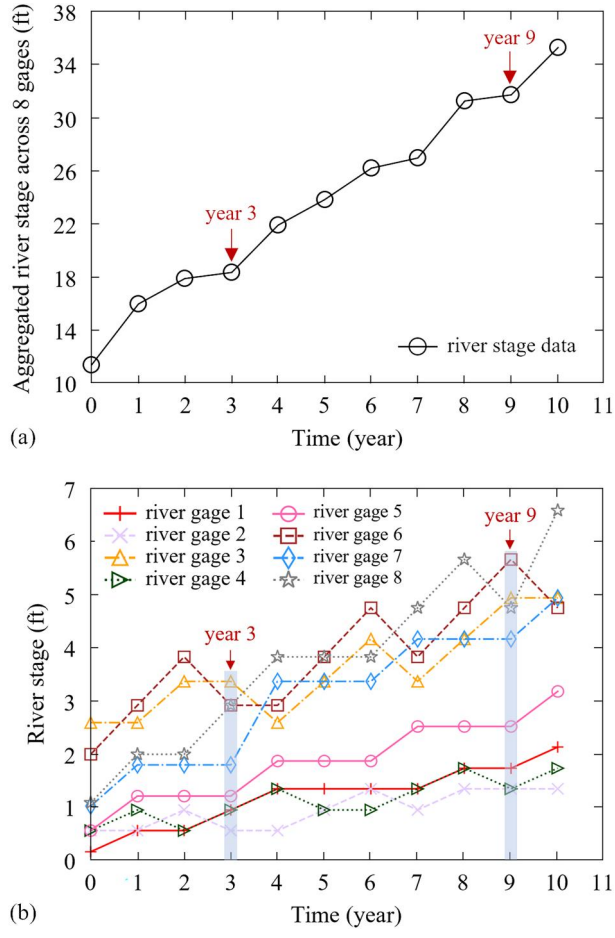


Figure 17. Hypothetical river stage data: (a) aggregated river stage across 8 gages and (b) individual river stage from each gage.

7. Conclusions

This paper proposes a performance-oriented system digital twin (SDT) framework for the modeling and management of smart infrastructure systems. This framework lays out the architecture, methodology and steps in building a user-centric decision support system. It is proposed to store a

mutable knowledge graph to represent the infrastructure systems within the framework. Based on the user's objective, the performance measure and multiple data sources of the targeted system, the knowledge graph is restructured. An algorithm, *SelTrunk*, is proposed to select a custom trunk from the restructured graph. This trunk consisting of relevant nodes sets the boundary for building the SDT and has an appropriate size to be handled with the computation expense limit. The trunk is then grown, and the graph is turned into a Bayesian network (BN), which is further transformed into an SDT in support of Bayesian inference and the evolution of BN.

The performance-oriented SDT tackles three challenges in modeling/managing infrastructure assets/networks which constitute interdependent complex systems. First, the complex statistical correlations and dependencies within and across infrastructure systems are addressed by using the BN for tasks such as risk analysis and management. Second, the evolution of infrastructure systems over time is considered by transforming and updating the mutable knowledge graph stored in the SDT. Third, event-responsive reasonings with different decision-making scenarios are accomplished by establishing a custom SDT based on the user's inputs using the proposed *SelTrunk* algorithm. A hypothetical graph and a real-world scenario are presented to demonstrate the advantages and applicability of the proposed SDT architecture:

- Three decision-making scenarios are considered for the hypothetical graph. Specifically, three sets of performance measure node and data nodes are provided to produce three restructured graphs. Results show that the *SelTrunk* algorithm can select different trunks based on different inputs. This demonstrates the flexibility of the performance-oriented SDT to consider various decision-making scenarios. The selected trunks also have a relatively small size, which improves the computation and storage efficiency.
- Based on the first scenario, a data-driven system reliability analysis is illustrated using the hypothetical graph. The BN is first constructed using the custom trunk and is then transformed into an SDT. Two cases (i.e. with and without one data source) are compared. The results show that the predicted failure probabilities differ when new relevant data sources are available. This indicates the generic flexibility of the proposed SDT for incorporating additional data sources as soon as they become available for improving systems modeling/management performance such as system reliability analysis.
- Another real-world case of the bridge network in Miami-Dade County is utilized to illustrate the practical use of the proposed framework. A large-scale knowledge graph is first created to represent the bridge network and relevant data nodes. Then, the network risk is added as the performance measure node and river gages are labeled as data nodes. Using the *SelTrunk* algorithm, the selected custom SDT has a size of 1% of the knowledge graph. A future scenario is hypothesized to perform Bayesian inference. It shows that the risk of the bridge network exhibits a consistent trend with river gage data, demonstrating the applicability and effectiveness of the proposed framework.
- The proposed framework suffers from the following limitations. While the current framework embeds the geometry of city/county into the KG, the geometry of individual assets at the component level is not handled. Future work should incorporate BIM standards to address this limitation. Additionally, the current framework primarily focuses on the computational aspects of DT. It does not delve into other modeling enabling technologies such as sensors, simulations, data-driven models, model predictive control, among others. Future work should integrate the proposed framework with these modeling enabling technologies to develop a comprehensive platform.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

Some or all data, models, or code that support the findings of this study are available from the first author upon reasonable request.

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